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Subject: Data Analytics with R

Experiment No. 02

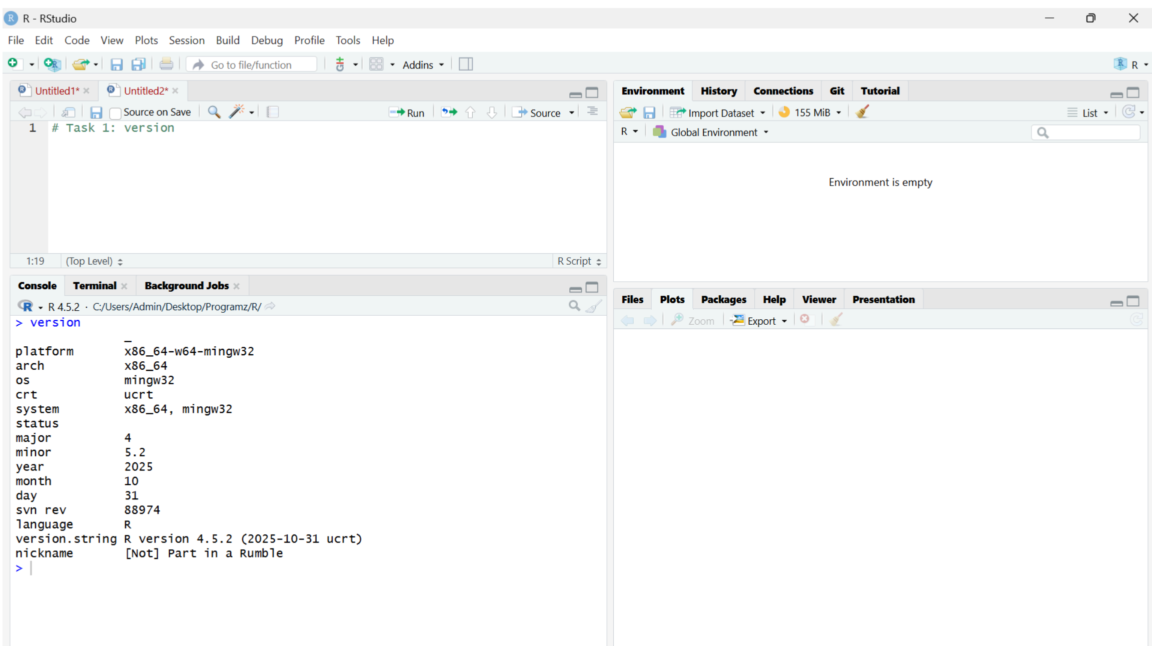
To install R and RStudio and create vector, matrices, arrays and factors and perform basic arithmetic operations and logical comparisons of R objects

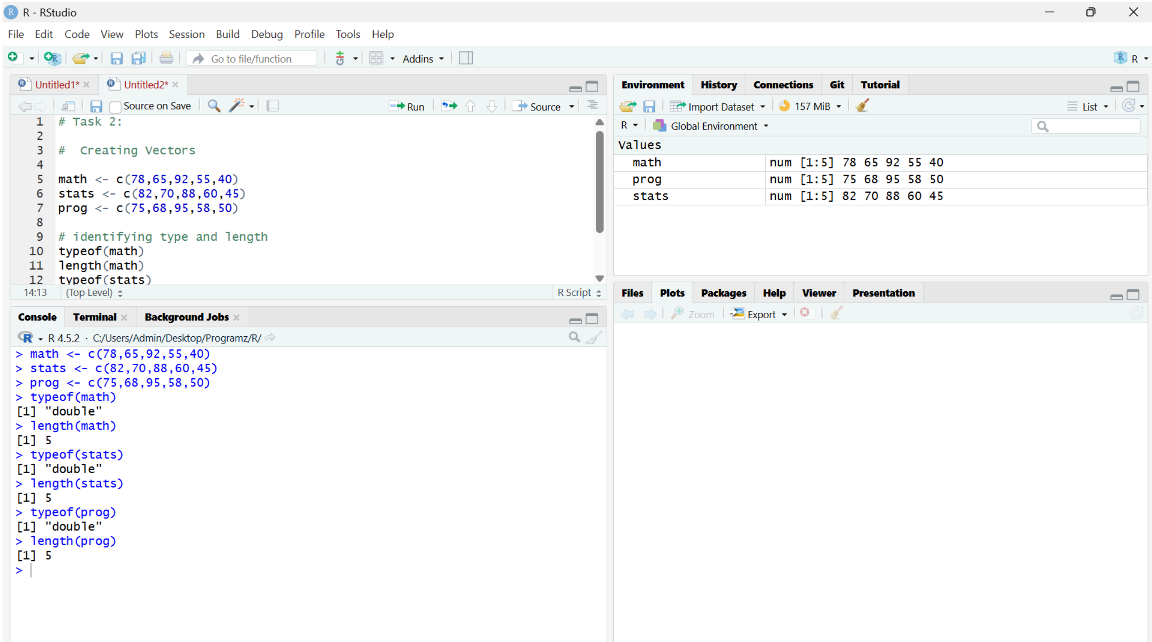
Title: To install R and RStudio and create vectors, matrices, arrays and factors and perform basic arithmetic operations and logical comparisons on R objects

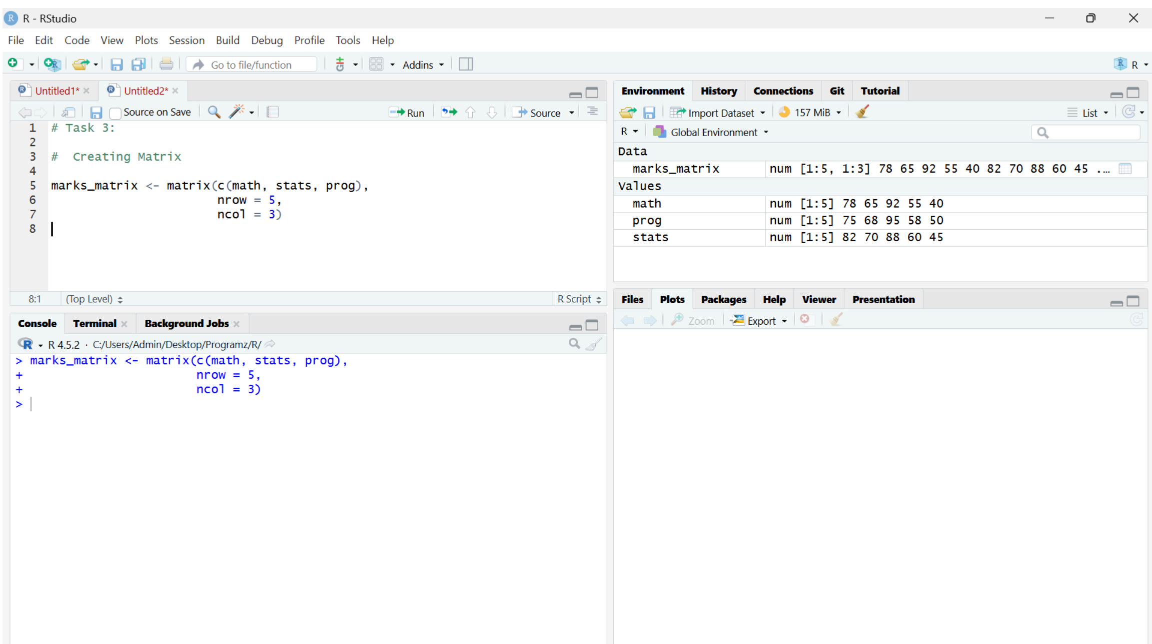
Aim:

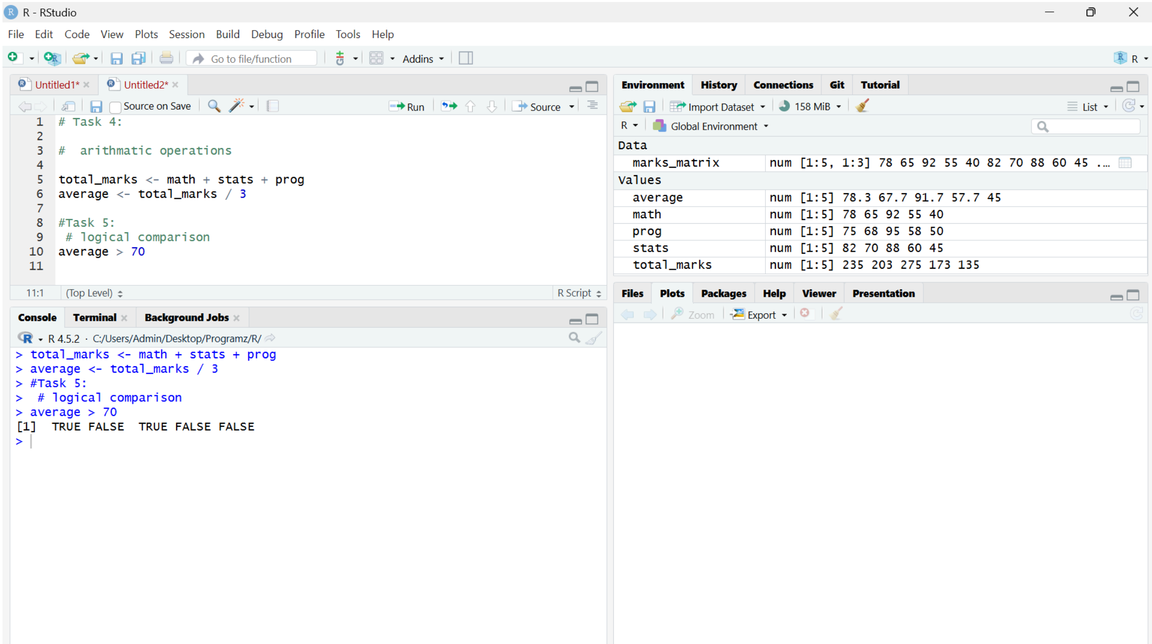
To install and configure the R programming environment and to understand fundamental R data structures such as vectors, matrices, arrays and factors along with performing arithmetic and logical operations on R objects

Code / Outputs:




R - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
Source on Save
Run
Source
# Task 2:
# Creating Vectors
math <- c(78,65,92,55,40)
stats <- c(82,70,88,60,45)
prog <- c(75,68,95,58,50)
# identifying type and length
typeof(math)
length(math)
typeof(stats)
14:13 (Top Level)
R Script
Console Terminal Background Jobs
R 4.5.2 · C:/Users/Admin/Desktop/Programz/R/
> math <- c(78,65,92,55,40)
> stats <- c(82,70,88,60,45)
> prog <- c(75,68,95,58,50)
> typeof(math)
[1] "double"
> length(math)
[1] 5
> typeof(stats)
[1] "double"
> length(stats)
[1] 5
> typeof(prog)
[1] "double"
> length(prog)
[1] 5
Environment History Connections Git Tutorial
Import Dataset
157 MiB
List
Global Environment
Values
math num [1:5] 78 65 92 55 40
prog num [1:5] 75 68 95 58 50
stats num [1:5] 82 70 88 60 45
Files Plots Packages Help Viewer Presentation
Zoom
Export


R - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
Source on Save
Run
Source
# Task 3:
# Creating Matrix
marks_matrix <- matrix(c(math, stats, prog),
nrow = 5,
ncol = 3)
8:1 (Top Level)
R Script
Console Terminal Background Jobs
R 4.5.2 · C:/Users/Admin/Desktop/Programz/R/
> marks_matrix <- matrix(c(math, stats, prog),
+ nrow = 5,
+ ncol = 3)
Environment History Connections Git Tutorial
Import Dataset
157 MiB
List
Global Environment
Data
marks_matrix num [1:5, 1:3] 78 65 92 55 40 82 70 88 60 45 ...
Values
math num [1:5] 78 65 92 55 40
prog num [1:5] 75 68 95 58 50
stats num [1:5] 82 70 88 60 45
Files Plots Packages Help Viewer Presentation
Zoom
Export


R - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Untitled2*
Source on Save
Run
Source
# Task 4:
# arithmatic operations
total_marks <- math + stats + prog
average <- total_marks / 3
#Task 5:
 # logical comparison
average > 70
11:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · C:/Users/Admin/Desktop/Programz/R/
> total_marks <- math + stats + prog
> average <- total_marks / 3
> #Task 5:
> # logical comparison
> average > 70
[1] TRUE FALSE TRUE FALSE FALSE
Environment
History
Connections
Git
Tutorial
Import Dataset
158 MiB
List
Global Environment
Data
marks_matrix num [1:5, 1:3] 78 65 92 55 40 82 70 88 60 45 ...
Values
average num [1:5] 78.3 67.7 91.7 57.7 45
math num [1:5] 78 65 92 55 40
prog num [1:5] 75 68 95 58 50
stats num [1:5] 82 70 88 60 45
total_marks num [1:5] 235 203 275 173 135
Files
Plots
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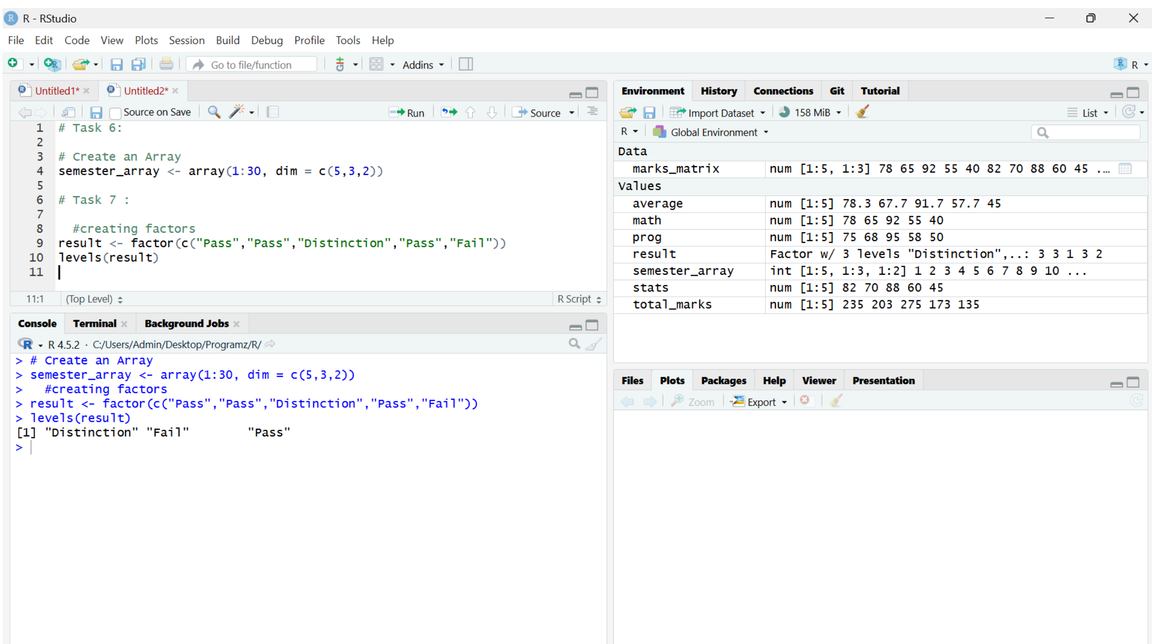
Help
Viewer
Presentation
Zoom
Export


R - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Untitled2*
Source on Save
Run
Source
# Task 6:
# Create an Array
semester_array <- array(1:30, dim = c(5,3,2))
# Task 7 :
  #creating factors
result <- factor(c("Pass","Pass","Distinction","Pass","Fail"))
levels(result)
11:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · C:/Users/Admin/Desktop/Programz/R/
> # Create an Array
> semester_array <- array(1:30, dim = c(5,3,2))
>   #creating factors
> result <- factor(c("Pass","Pass","Distinction","Pass","Fail"))
> levels(result)
[1] "Distinction" "Fail"   "Pass"
Environment
History
Connections
Git
Tutorial
Import Dataset
158 MiB
List
Global Environment
Data
marks_matrix num [1:5, 1:3] 78 65 92 55 40 82 70 88 60 45 ...
Values
average num [1:5] 78.3 67.7 91.7 57.7 45
math num [1:5] 78 65 92 55 40
prog num [1:5] 75 68 95 58 50
result Factor w/ 3 levels "Distinction",..: 3 3 1 3 2
semester_array int [1:5, 1:3, 1:2] 1 2 3 4 5 6 7 8 9 10 ...
stats num [1:5] 82 70 88 60 45
total_marks num [1:5] 235 203 275 173 135
Files
Plots
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Help
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Presentation
Zoom
Export

Observation Table:

Task No	R Operation	R Code Used	Result Observed	Observation
1	Installation Verification	`version`	Displays R version and platform details	R and RStudio installed successfully
2	Vector Creation (Math)	`math <- c(78, 65, 92, 55, 40)`	Numeric vector of length 5	Vector stores homogeneous data
2	Vector Creation (Stats)	`stats <- c(82, 70, 88, 60, 45)`	Numeric vector of length 5	All elements are numeric
2	Vector Creation (Prog)	`prog <- c(75, 68, 95, 58, 50)`	Numeric vector of length 5	Length = 5
3	Matrix Creation	`marks_matrix <- matrix(c(math, stats, prog), nrow=5, ncol=3)`	5×3 numeric matrix	Data organized in rows & columns

Conclusion:

The experiment successfully introduced the R programming environment and fundamental data structures used in data analytics. Students learned to create vectors, matrices, arrays and factors and performed arithmetic and logical operations. These concepts form the foundation for data manipulation and analysis in subsequent experiments.